

S. Shayan Daneshvar

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PROFESSIONAL SUMMARY

LLM research engineer specializing in post-training for agentic coding: verifiable SWE environments, rejection-sampled distillation, SFT, and multi-node inference and evaluation for dense/MoE models, including the open-source Pangu 92B. First-author papers in ACM TOSEM (Q1) and EASE (CORE A), plus strong production software engineering experience.

TECHNICAL SKILLS

LLM Post-Training: SFT, Rejection Sampling, Knowledge Distillation, DPO, RLVR, LoRA

Distributed Training & Inference: Megatron-LM, LLaMA-Factory, DeepSpeed ZeRO, 3D Parallelism, vLLM

AI Agents & Evaluation: LangGraph, LangChain, MCP, Agentic RAG, Harbor, Weave, G-Eval, Claude Code, Codex

Machine Learning & Deep Learning: PyTorch, Hugging Face Transformers, OpenCV, WandB, Computer Vision, NLP, Domain Adaptation, Data Augmentation, Diffusion Models, Synthetic Data Generation, Contrastive Learning

Programming Languages: Python, Java, SQL, Bash

Software Engineering: Spring Boot, Microservices, Docker, Kubernetes, Kafka, NoSQL, CI/CD, AWS, Scrum/Agile

PROFESSIONAL EXPERIENCE

LLM Research Engineer

Nov 2025 – present

Huawei Technologies Canada

Kingston, Canada

- Proposed and built a sandbox-reusing task augmentation pipeline that grew rejection-sampled distillation data by 50%, lifting Qwen3-14B via SFT by **4.3 pp on SWE-bench Verified** and **7.1 pp on FeatBench**
- Architected an agentic SWE environment builder whose **procedural memory** evolves into an **executable ruleset**, building 10K+ environments across 10 languages; vs. SWE-Gen: **2.2× yield**, **3.8× throughput**, 68% fewer tokens
- Designed multi-stage guidance for **agent evaluation** to reduce false failures from hidden API mismatches, yielding **fairer benchmarking** and graded training signals for PPO/DPO
- Developed a trajectory quality analyzer with integrated agent tool call classification by phase to flag loops, brute-forcing, and skipped verification, marking **12%** of issue-resolution trajectories for redistillation
- Performed continual **supervised fine-tuning** of **Pangu 92B MoE** on 1B+ tokens of my distilled trajectories combined with a subsample of its original 15B-token code data, across multi-node distributed GPU/NPU clusters

Machine Learning Engineer

Apr 2025 – Sep 2025

MacDon Industries (Contract via Zoomi Technologies)

Winnipeg, Canada

- Led end-to-end delivery of a DAFormer-based unsupervised domain adaptation model with a custom ClassMix, beating the supervised baseline by **3 pp** in Dice score using only unlabelled real data, in 75% of the planned time
- Customized DoGE generative augmentation by conditioning **ControlNet** on filtered masks and edges; cut labelling real data requirement by **70%** while maintaining semantic segmentation quality

Machine Learning Engineer Intern

Nov 2024 – Mar 2025

MacDon Industries (Mitacs Accelerate)

Winnipeg, Canada

- Integrated and trained SegFormer in the legacy PyTorch pipeline for a **4 pp Dice** gain, and diagnosed real-vs-synthetic drift with **UMAP** on **CLIP** and **VGG embeddings**, steering the project to domain adaptation

Graduate Research Assistant

Jan 2023 – Mar 2025

University of Manitoba

Winnipeg, Canada

- First-authored VulScribeR (**TOSEM**), the first use of **LLMs** for vulnerability data augmentation, with **RAG**-guided generation lifting detector F1 by **~28%** (relative) over SOTA generators at US\$1.88 per 1K samples
- First-authored the first systematic study of mixup-based augmentation for vulnerability detection (**EASE**)

Senior Software Engineer

May 2022 – Dec 2022

Mahsan

Tehran, Iran

- Guided a **team of 4** in architecting a distributed data-analytics platform for dynamic ontology management, combining **microkernel** and **microservices** architectures with **CQRS**, **Saga**, and **Event Sourcing**
- Built **distributed tracing** and **monitoring** for all microservices with Spring Cloud Sleuth, Zipkin, Prometheus, Grafana, Elasticsearch, Kibana, and Logstash
- Led 50+ code reviews enforcing coding standards, mentored 2 junior developers, and taught Spring Cloud and microservices to **7+ developers** over 30+ hours

Backend Software Engineer

Nov 2021 – May 2022

Tosan Soha

Tehran, Iran

- Developed hybrid Java/Kotlin e-wallet and trading platforms on **Kubernetes** serving **2K+ concurrent users**, with banking APIs for internal and external services and event-driven Kafka workflows
- Proposed replacing a tangled JWT/session login system with **Spring Security OAuth**-based client/server services to enable third-party login and implemented parts of the OAuth flow as well as cross-service refactorings

Java Backend Developer

May 2021 – Nov 2021

Pinket

Tehran, Iran

- Developed event-driven chat, notification, and shopping services with Spring, Kafka, PostgreSQL, and MongoDB
- Maintained the chat service and added integration and unit tests to CI/CD, raising code coverage by **7%**

Java Instructor and Mentor (Part-time)

Jan 2020 – May 2021

Mapsa HR and Training

Tehran, Iran

- Taught core and advanced Java, including OOP, SQL, data structures, concurrency, and algorithm design
- Delivered hands-on training and mentorship in Spring Boot, Spring MVC, JPA, Docker, and REST APIs

EDUCATION

University of Manitoba

Winnipeg, Canada

Master of Science in Computer Science, Deep Learning for Software Engineering, 4.5/4.5 GPA

- **Thesis:** Exploring Representation-level Augmentation and RAG-based Vulnerability Augmentation with LLMs for Vulnerability Detection
- **Supervisor:** Dr. Shaowei Wang

K. N. Toosi University of Technology

Tehran, Iran

Bachelor of Science in Computer Engineering, 4/4 GPA, ranked 1st of 76

- **Thesis:** Reflection Removal of In-vehicle Images with UNets
- **Advisor:** Dr. Behrooz Nasihatkon

RESEARCH PUBLICATIONS

VulScribeR: Exploring RAG-based Vulnerability Augmentation with LLMs 2026

S. Daneshvar, Y. Nong, X. Yang, S. Wang, H. Cai – ACM TOSEM (Q1, top software engineering journal)

A Study on Mixup-Inspired Augmentation Methods for Software Vulnerability Detection 2025

S. Daneshvar, D. Tan, S. Wang, C. Leung – EASE 2025 (CORE A)

GUI Element Detection Using SOTA YOLO Deep Learning Models 2024

S. Daneshvar, S. Wang – arXiv preprint

The application of barcode readable assay and linear regression RGB analysis... 2024

N. Rezaei, S. Daneshvar, B. Nasihatkon, S. Seidi, M. Rezazadeh – Microchemical Journal (Q1)

HONORS & SERVICE

Service: Reviewer, ACM TOSEM (2026–present) • ICSE 2026 Shadow Program Committee

Awards: UManitoba Graduate Fellowship • Research Completion Award • International Graduate Entrance Scholarship

Distinctions: Ranked 1,427th of 144,437 (top 1%) in Iran's national university entrance exam

PROJECTS

Reflection Removal with Mamba/S6 State-Space Model | Python, Docker, OpenCV, PyTorch 2024

- Built ImageMamba, an adapter that runs the original Mamba state-space module on image feature maps, and replaced DSRNet's attention modules with it, beating the SOTA DSRNet on PSNR (**+0.14 dB**) and SSIM

Chibaladi.com | Spring Cloud, PostgreSQL, MongoDB, Spring Data, Nginx 2021

- Designed, built, and deployed a microservices-based e-learning platform with video courses and adaptive quizzing, independently owning backend architecture, development, and deployment end-to-end

LANGUAGES

English (Fluent, C1 – CELPIP 11/12) • Persian (Native)